

Laplace Analysis and Applications

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1 From Functions to Transforms

1.1 The Laplace Transform

Let $f(t)$ be a function defined for $t \geq 0$. Its Laplace transform is defined by

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt, \quad s > 0. \quad (1)$$

The transform changes a function of time into a function of the new variable s .¹ Symbolically,

$$f(t) \xrightarrow{\mathcal{L}} F(s).$$

The inverse Laplace transform reverses this process:

$$f(t) = \mathcal{L}^{-1}\{F(s)\}. \quad (2)$$

The main purpose of the method is to replace differentiation and integration in the time domain by simpler algebraic operations in the transform domain [1]. A concise overview is available from the [Laplace Transform entry on MathWorld](#).

1.2 A Simple Example

For the constant function $f(t) = 1$,

$$\begin{aligned} \mathcal{L}\{1\} &= \int_0^{\infty} e^{-st} dt \\ &= \left[-\frac{1}{s} e^{-st} \right]_0^{\infty} \\ &= \frac{1}{s}. \end{aligned} \quad (3)$$

Similarly,

$$\mathcal{L}\{t\} = \frac{1}{s^2}, \quad \mathcal{L}\{t^2\} = \frac{2}{s^3}. \quad (4)$$

2 Common Transform Equations

Table 1 lists several frequently used Laplace-transform equations.

The formulas in Table 1 allow many functions to be transformed without repeatedly evaluating improper integrals.

¹The variable s is commonly interpreted as a complex-frequency variable.

Table 1: Selected Laplace-transform equations.

Family	Transform Equation		Condition
	Time function	Laplace transform	
Algebraic	1	$\frac{1}{s}$	$s > 0$
	t	$\frac{1}{s^2}$	$s > 0$
	t^n	$\frac{n!}{s^{n+1}}$	$n = 0, 1, 2, \dots$
Exponential	e^{at}	$\frac{1}{s-a}$	$s > a$
	e^{-at}	$\frac{1}{s+a}$	$s > -a$
Oscillatory	$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$	$\omega > 0$
	$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$	$\omega > 0$
Unit-step function $u(t-a)$		$\frac{e^{-as}}{s}$	$a > 0$

3 Application to an RC Circuit

Circuit Model

Figure 1 shows a simple **resistor--capacitor circuit** connected to an input voltage source. The resistor limits the current, while the capacitor stores electrical energy.

For a resistor R and capacitor C , Kirchhoff's voltage law gives

$$RC \frac{dv_C(t)}{dt} + v_C(t) = v_{in}(t), \quad (5)$$

where $v_C(t)$ is the voltage across the capacitor. Assuming $v_C(0) = 0$, taking the Laplace transform of Equation (5) gives

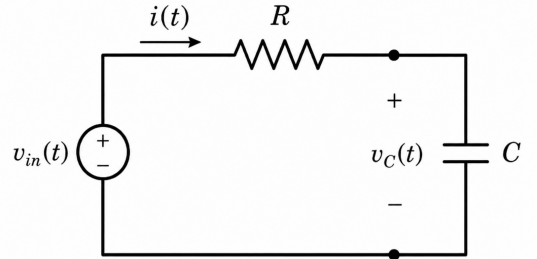


Figure 1: A simple series RC circuit.

$$\begin{aligned}
 RC [sV_C(s) - v_C(0)] + V_C(s) &= V_{in}(s), \\
 \Rightarrow RCsV_C(s) + V_C(s) &= V_{in}(s), \\
 \Rightarrow V_C(s)(RCs + 1) &= V_{in}(s).
 \end{aligned} \quad (6)$$

The transfer function is therefore

$$H(s) = \frac{V_C(s)}{V_{in}(s)} = \frac{1}{RCs + 1}. \quad (7)$$

For a constant input V_0 ,

$$V_{in}(s) = \frac{V_0}{s}.$$

Thus,

$$V_C(s) = \frac{V_0}{s(RCs + 1)}. \quad (8)$$

Taking the inverse transform gives

$$v_C(t) = V_0 \left(1 - e^{-t/(RC)}\right). \quad (9)$$

The quantity $\tau = RC$ is called the time constant of the circuit. A larger value of τ causes the capacitor to charge more slowly [2].

4 How Laplace Analysis Is Used

The Laplace transform is useful in many areas:

Differential equations $\oplus$ initial conditions are included automatically;

$\oplus$ derivatives become algebraic expressions.

Electrical circuits $\oplus$ voltages and currents are studied in the s -domain;

$\oplus$ components lead to transfer functions.

Control and signal systems $\oplus$ stability can be studied through poles;

$\oplus$ common test inputs include:

▷ the unit-step input;

▷ the impulse input;

▷ sinusoidal inputs.

In the time domain, a system may be described by a differential equation, whereas in the Laplace domain, the same system is often described by a rational algebraic expression.

5 A Short Python Implementation

Listing 1 evaluates the capacitor voltage at a finite set of time instants using an external Python file.

Listing 1: Computing samples of the RC step response.

```
1 from math import exp
2
3
4 def rc_response(resistance, capacitance, input_voltage, times):
5     """Return capacitor-voltage samples for an RC step input."""
6     if resistance <= 0 or capacitance <= 0:
7         raise ValueError("R and C must be positive")
8
9     tau = resistance * capacitance
10    values = []
11
12    for time in times:
13        voltage = input_voltage * (1 - exp(-time / tau))
14        values.append(voltage)
15
16    return values
17
18
19 times = [0.0, 0.5, 1.0, 2.0]
20 print(rc_response(2.0, 0.5, 5.0, times))
```

Equation (9) and Listing 1 describe the same computation.

References

- [1] E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed. John Wiley & Sons, 2011.
- [2] K. Ogata, *Modern Control Engineering*, 5th ed. Prentice Hall, 2010.